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VHFACSerialBusDefTopic



About the VHFAC Serial Bus

About the VHFAC Serial Bus

Each VHFAC board includes a serial bus which generates serial digital information useful for register setting and I²C control.

VHFAC Serial Bus information includes the following sections:

- [VHFAC Serial Bus Theory of Operation](#)

- [VHFAC Serial Bus Programming](#)

- [VHFAC Serial Bus Debug Display](#)

- [VHFAC Serial Bus Examples](#)

Related Topics

- [VHFAC Specifications](#)

- [VHFACSerialBus](#) (VBT syntax)

- [VHFAC DIB Design Guidelines](#)

For information on other VHFAC instruments see:

- [About the VHFAC Capture](#)

- [About the VHFAC Source](#)

- [About the Time Measurement Unit](#)

- [About the VHFAC Trigger Control](#)

Serial Bus Characteristics

The serial bus characteristics listed in the table provide a guide to its capabilities. For hardware specifications, see VHFAC Specifications.

Digital Serial Bus for Register Setting and I²C Control

Characteristic	Condition	Value	Units	Notes
Clock Channel		1		
Drive/Receive		2		
Vector Rate		0.0024 – 1.67	MHz	
Voltage Levels	No Load	(-4) –10	V	
Edge Placement Accuracy		30	ns	Displacement between two channels programmed to the same point in time
Pattern Memory		64	kVectors	

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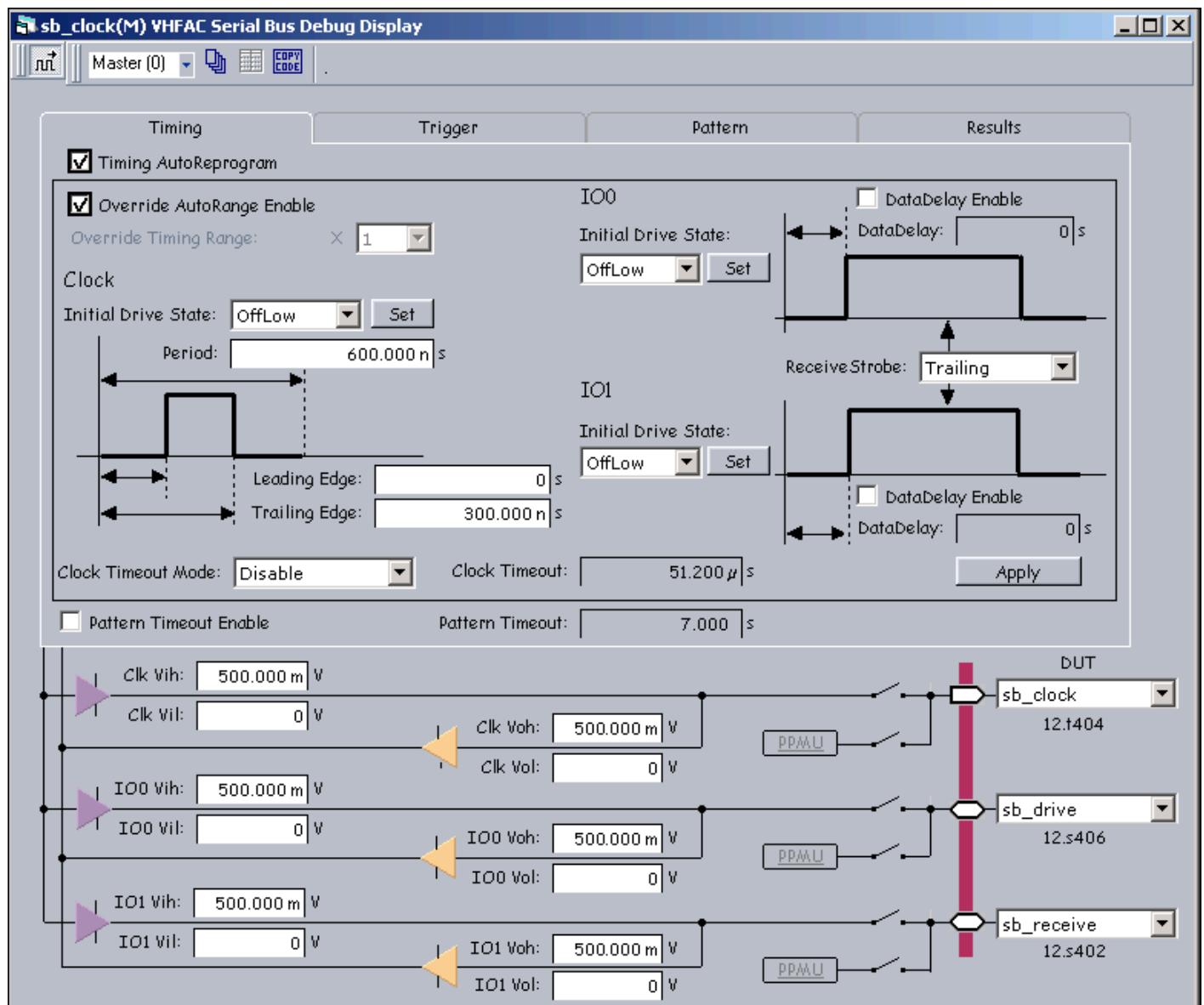
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VHFAC Serial Bus Debug Display

VHFAC Serial Bus Debug Display

The VHFAC Serial Bus Debug Display is an interactive tool that shows the state of the VHFAC Serial Bus instrument in real time. Most of the parameters are interactive. Features that can be changed may be limited to a certain range or selection of options provided in a drop-down menu. Non-interactive features are grayed out on the display and cannot be adjusted during debugging. This section describes features specific to the VHFAC Serial Bus debug display. For more information on using debug displays, see the following:

- | | |
|---|--|
| • | Using Debug Displays in TDE |
| • | Producing Code from Debug Displays |



Serial Bus Debug Display

Starting the VHFAC Serial Bus Debug Display

To launch the VHFAC Serial Bus Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>VHFAC Serial Bus.

To start debugging, select a pin and a site to display the programmed values. Select sites and pins from the drop-down menu.

Without a test program loaded, the initial debug display returns an error:

! No pin exists of type VHFAC Serial Bus

and the connection lines appear directly connected to the DUT (without relays, DIB Access, or PPMU instruments active). Load a validated test program and restart the debug display to remedy this situation.

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VHFAC Serial Bus Debug Display : Debug Display Parameters and Buttons

Debug Display Parameters and Buttons

The VHFAC Serial Bus Debug Display contains several sections. Each section has programmable features used to make adjustments to test parameters. The sections are:

- VHFAC Serial Bus Timing
- VHFAC Serial Bus Pin Levels
- VHFAC Serial Bus Pattern

Pattern Data

Pattern Set

Pattern Group
- VHFAC Serial Bus Results
- VHFAC Serial Bus Trigger

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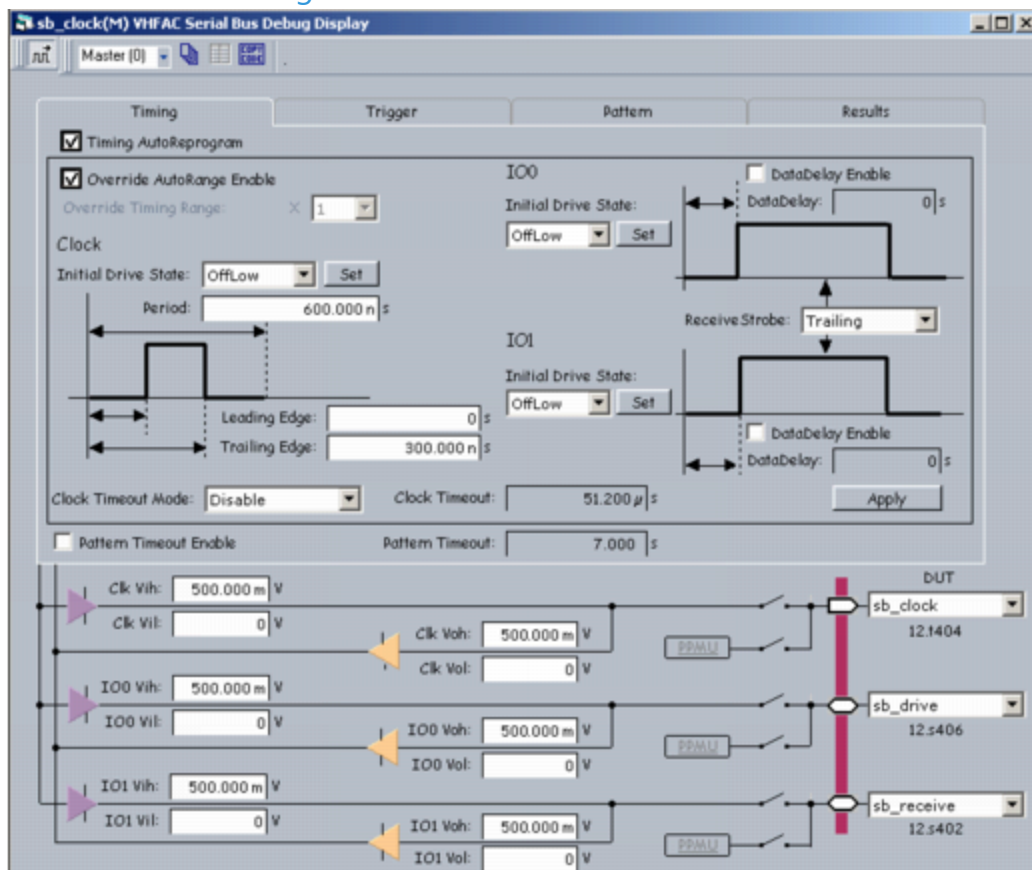
VhfacSerialBus_debug.4.3



VHFAC Serial Bus Debug Display : Debug Display Parameters and Buttons : VHFAC Serial Bus Timing

VHFAC Serial Bus Timing

The VHFAC Serial Bus Timing section displays the current programmed settings of the selected VHFAC Serial Bus instrument and allows changes to be made to program parameters. The timing tab is shown in the figure.



Serial Bus Debug Display Timing Tab

The controls for the Timing tab are as follows.

VHFAC Serial Bus Schematic: Timing Tab Controls

Parameters/

Buttons

Description

Timing AutoReprogram	This check-box shows the state of Timing AutoReprogram enable. This property enables or disables automatic reprogramming of all the other timing properties when the range changes. • Selected: Timing AutoReprogram Enabled • Cleared: Timing Auto Reprogram Disabled
Override AutoRange Enable	This check-box is selected to set up a timing range automatically from a period. • Selected: Timing AutoReprogram Enabled • Cleared: Timing Auto Reprogram Disabled
Override Timing Range	This drop-down list manually programs the override timing range. The resolution for the VHFAC Serial Bus can be control by setting this parameter to one of the timing ranges: 1, 2, 4, 8, or 16. This control is grayed out if Override AutoRange Enable is selected.
Period	This text-box specifies the length of the cycle and can be from 600 ns to 409.6s. The reset state is 600 ns. The VHFAC Serial Bus has timing registers with the LSB time setting of 100 ns. The minimum period is $(6 * 100\text{ns} * \text{timing range})$. With a timing range of x1, the minimum period is 600 ns. The maximum period is $(256 * 100\text{ns} * \text{timing range}) = 25.6\text{s}$ with a timing range of x1. The length of the period can be extended beyond 25.6s by increasing the timing range. However, increasing the time setting also decreases the timing resolution. The longer period is achieved by multiplying the amount of time the LSB times of 100ns, 200ns, 400ns, 800ns, or 1.6s. With the LSB time multiplied by 16, the period can reach its maximum of 409.6s.

Leading Edge	This text-box specifies the placement of the clock leading edge within the clock cycle. Minimum value is 0. Maximum value is $(\text{Period} - \text{timing range} * 100\text{ns})$.
Trailing Edge	This property specifies the placement of the clock trailing edge within the clock cycle. Minimum value is $(\text{LeadingEdge} + \text{timing range} * 100\text{ns})$. Maximum value is $(\text{Period} - \text{timing range} * 100\text{ns})$.
Clock Timeout Mode	This drop-down list provides a method of setting the maximum amount of time the hardware waits for devices to release the clock line. • Disable: Clock Timeout Mode is disabled • Leading: Leading edge of the clock is used for rising edge • Trailing: Trailing edge of the clock is used for rising edge
Clock Timeout	This text-box specifies the value of clock timeout. The range is from 200ns to $(\text{timing range} * 256 * 200\text{ns})$.
DataDelay Enable	This check-box shows the state of DataDelay enable: • Selected: DataDelay Enabled • Cleared: DataDelay Disabled
DataDelay	This text-box specifies a timing delay. Each channel can begin its timing at the start of the period or after the DataDelay. The range is from 0 to $(\text{Period} - \text{timing range} * 100\text{ns})$.
Receive Strobe	This drop-down list selects the leading edge or trailing edge of the clock as the strobe for the receiver channels. • Leading • Trailing
Initial Drive State	This drop-down list sets the initial drive state for the clock and I/O channels.

- **OffLow**: Sets driver voltage level to Vil, but driver is turned off.
- **OffHigh**: Sets driver voltage level to Vih, but driver is turned off.
- **Low**: Sets driver voltage level to Vil.
- **High**: Sets driver voltage level to Vih.

Pattern Timeout Enable

This check-box shows the state of pattern timeout enable, and provides a method of setting the maximum time the test program waits for a pattern to finish execution.

- **Selected**: Pattern Timeout Enabled
- **Cleared**: Pattern Timeout Disabled

Pattern Timeout

This text-box specifies the value of pattern timeout. The range is from 100ns to 7s.

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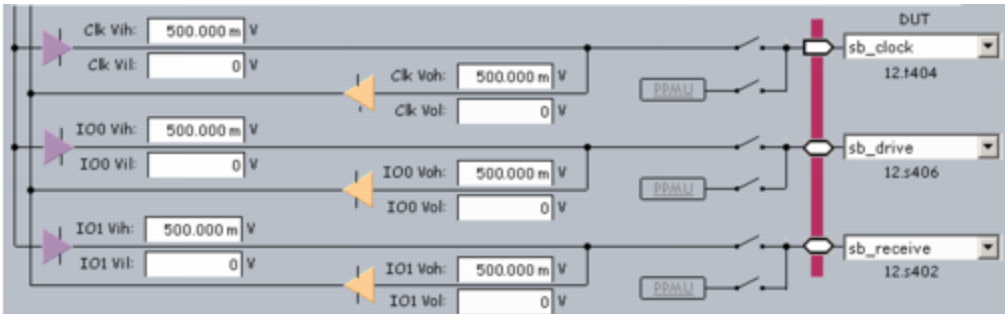
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VHFAC Serial Bus Debug Display : Debug Display Parameters and Buttons : VHFAC Serial Bus Pin Levels

VHFAC Serial Bus Pin Levels

Pin Levels are shown below the tabs of the debug display.



Serial Debug Display, Pin Levels

The controls for this part of the display are as follows:

VHFAC Serial Bus Schematic: Pin Levels Controls

Parameters/ Buttons	Description
Clk Vih	This text box specifies drive high voltage level of clock line. The range is from -4.0V to 10.0V.
Clk Vil	This text box specifies drive low voltage level of clock line. The range is from -4.0V to 10.0V.
Clk Voh	This text box specifies high comparator voltage level of clock line. The range is from -2.0V to 7.5V.
Clk Vol	This text box specifies low comparator voltage level of clock line. The range is from -2.0V to 7.5V.

IO0 Vih	This text box specifies drive high voltage level of IO0 data line. The range is from -4.0V to 10.0V.
IO0 Vil	This text box specifies drive low voltage level of IO0 data line. The range is from -4.0V to 10.0V.
IO0 Voh	This text box specifies high comparator voltage level of IO0 data line. The range is from -2.0V to 7.5Vv.
IO0 Vol	This text box specifies low comparator voltage level of IO0 data line. The range is from -2.0V to 7.5V.
IO1 Vih	This text box specifies drive high voltage level of IO1 data line. The range is from -4.0V to 10.0V.
IO1 Vil	This text box specifies drive low voltage level of IO1 data line. The range is from -4.0V to 10.0V.
IO1 Voh	This text box specifies high comparator voltage level of IO1 data line. The range is from -2.0V to 7.5V.
IO1 Vol	This text box specifies low comparator voltage level of IO1 data line. The range is from -2.0V to 7.5V.

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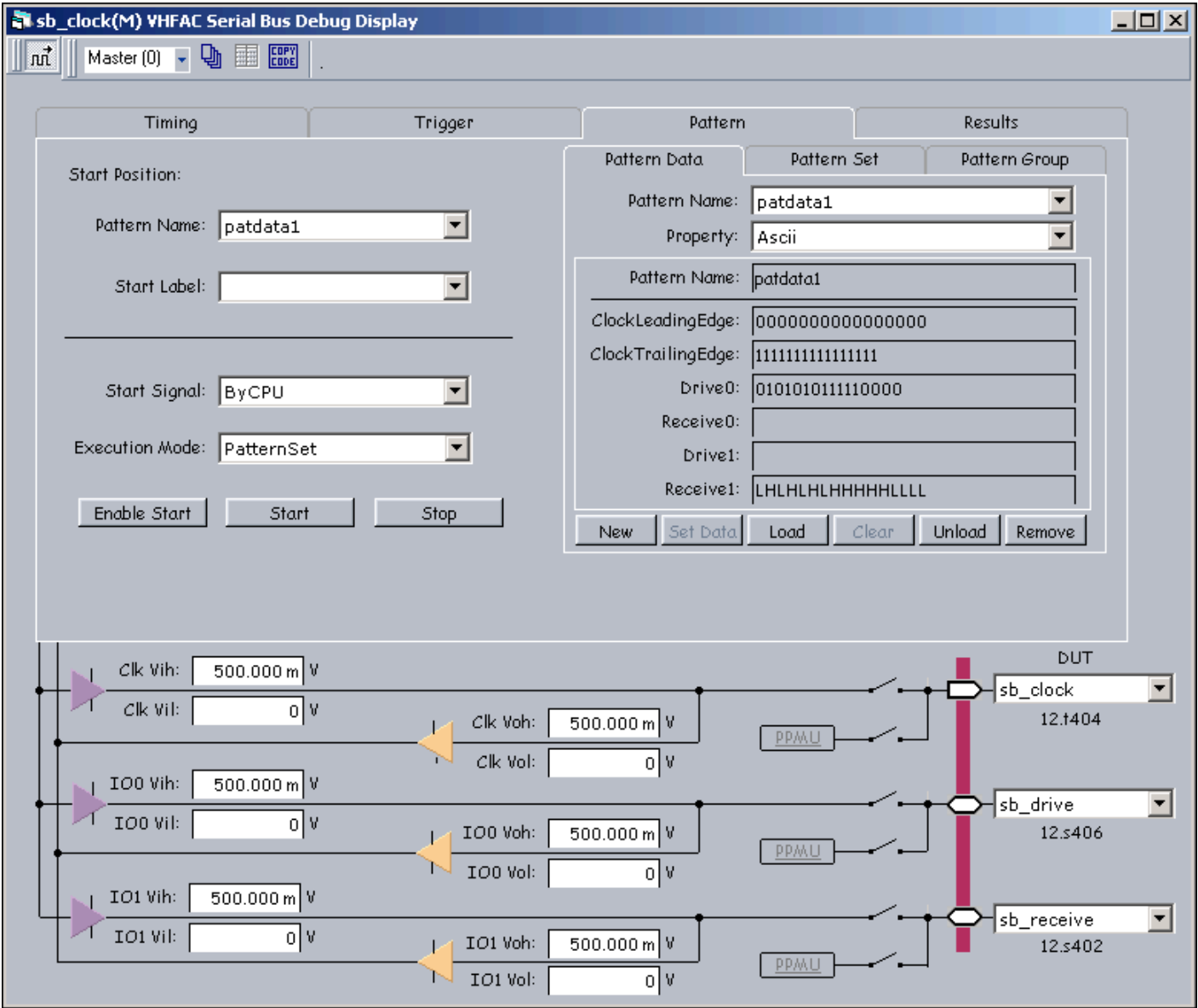
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[VHFAC Serial Bus Debug Display : Debug Display Parameters and Buttons](#) : VHFAC Serial Bus Pattern

VHFAC Serial Bus Pattern

When the Pattern tab is selected, a set of parameters becomes visible (on the left side of the debug display) and a set of Pattern subtabs becomes visible (on the right side of the display).



VHFAC Serial Bus Schematic: Pattern Group Tabs

The controls for the Pattern tab (left side) are as follows:

VHFAC Serial Bus Schematic: Pattern Tab Controls

Parameters/ Buttons	
	Description
Pattern Name	This drop-down list displays the loaded pattern name. A pattern name to perform can be chosen.
Start Label	This drop-down list displays the start label of the loaded pattern. The start label of a pattern to perform can be chosen.
Start Signal	This drop-down list selects the start trigger of a pattern.

- [ByCPU](#): Starting a pattern from test program
- [ByTrigger](#): Starting a pattern from trigger line

Execution Mode	This drop-down list can change into a pattern group temporarily the pattern defined by the pattern set. Usually, the pattern of pattern set is stopped in the end of the pattern data. The pattern can be stopped by changing execution mode in the end of the pattern. • PatternGroup: It performs to the end of pattern data. • PatternSet: It performs to the end of pattern
Enable Start	Sets starting position of pattern as hardware and initialize register.
Start	Manually starts the pattern for debug purposes.
Stop	Manually stops the pattern when being used for debug purposes.

Pattern Data

The Pattern Data tab (a subtab on the right side of the display) has the following controls:

VHFAC Serial Bus Schematic: Pattern Data Tab Controls

Parameters/ Buttons	Description
Pattern Name	This drop-down list displays the defined pattern name. The pattern data of the selected pattern can be displayed.
Property	This drop-down list selects the format of pattern data: Ascii, Hex, Bin, or Oct.
Pattern Name	This text box shows selected pattern data name. You can also input the pattern data name to add.
ClockLeadingEdge	This text box displays the clock leading edge data of the pattern data selected by the pattern name. You can also input the clock leading edge data to add.
ClockTrailingEdge	This text box displays the clock trailing edge data of the pattern data selected by the pattern name. You can also input the clock leading

	edge data to add.
Drive0	This text box displays the IO0 drive data of the pattern data selected by the pattern name. You can also input the IO0 drive data to add.
Receive0	This text box displays the IO0 receive data of the pattern data selected by the pattern name. You can also input the IO0 receive data to add.
Drive1	This text box displays the IO1 drive data of the pattern data selected by the pattern name. You can also input the IO1 drive data to add.
Receive1	This text box displays the IO1 receive data of the pattern data selected by the pattern name. You can also input the IO1 receive data to add.
Set Data	Adds the pattern data which the above parameter defined.
New	Adds a new pattern.
Set Data	Sets the data for a new pattern.
Load	Loads the pattern data selected by the pattern name to a pattern memory.
Clear	Clears the pattern data defined above.
Unload	Unloads the pattern data selected by the pattern name.
Remove	Removes the pattern data selected by the pattern name.

Pattern Set

The Pattern Set tab (a subtab on the right side of the display) has the following controls:

VHFAC Serial Bus Schematic: Pattern Set Tab Controls

Parameters/ Buttons	Description
Pattern Set	This drop-down list displays the defined pattern set name. The contents of the selected pattern set can be displayed.
Pattern Set	This text-box shows selected pattern set name. You can also input the pattern set name to add.
Pattern Name	This drop-down list displays the defined pattern data name. To add a pattern set, you can select a pattern data name.
Start Label	To add a pattern set, you can input a start label.
Repeat	To add a pattern set, you can input a repeat number.

Add Set	Adds the pattern set data which the above parameter defined.
Load	Loads the pattern set data selected by the pattern name to a pattern memory.
Unload	Unloads the pattern set data selected by the pattern name.
Remove	Removes the pattern set data selected by the pattern name.

Pattern Group

The Pattern Group tab (a subtab on the right side of the display) has the following controls:

VHFAC Serial Bus Schematic: Pattern Group Tab Controls

Parameters/ Buttons	Description
Pattern Group	This drop-down list displays the defined pattern group name. The contents of the selected pattern group can be displayed.
Pattern Group	This text-box shows selected pattern group name. You can also input the pattern group name to add.
Pattern Name	This drop-down list displays the defined pattern data name. To add a pattern group, you can select a pattern data name.
Repeat	To add a pattern group, you can input a repeat number.
Add Group	Adds the pattern group data which the above parameter defined.
Load	Loads the pattern group data selected by the pattern name to a pattern memory.
Unload	Unloads the pattern group data selected by the pattern name.
Remove	Removes the pattern group data selected by the pattern name.

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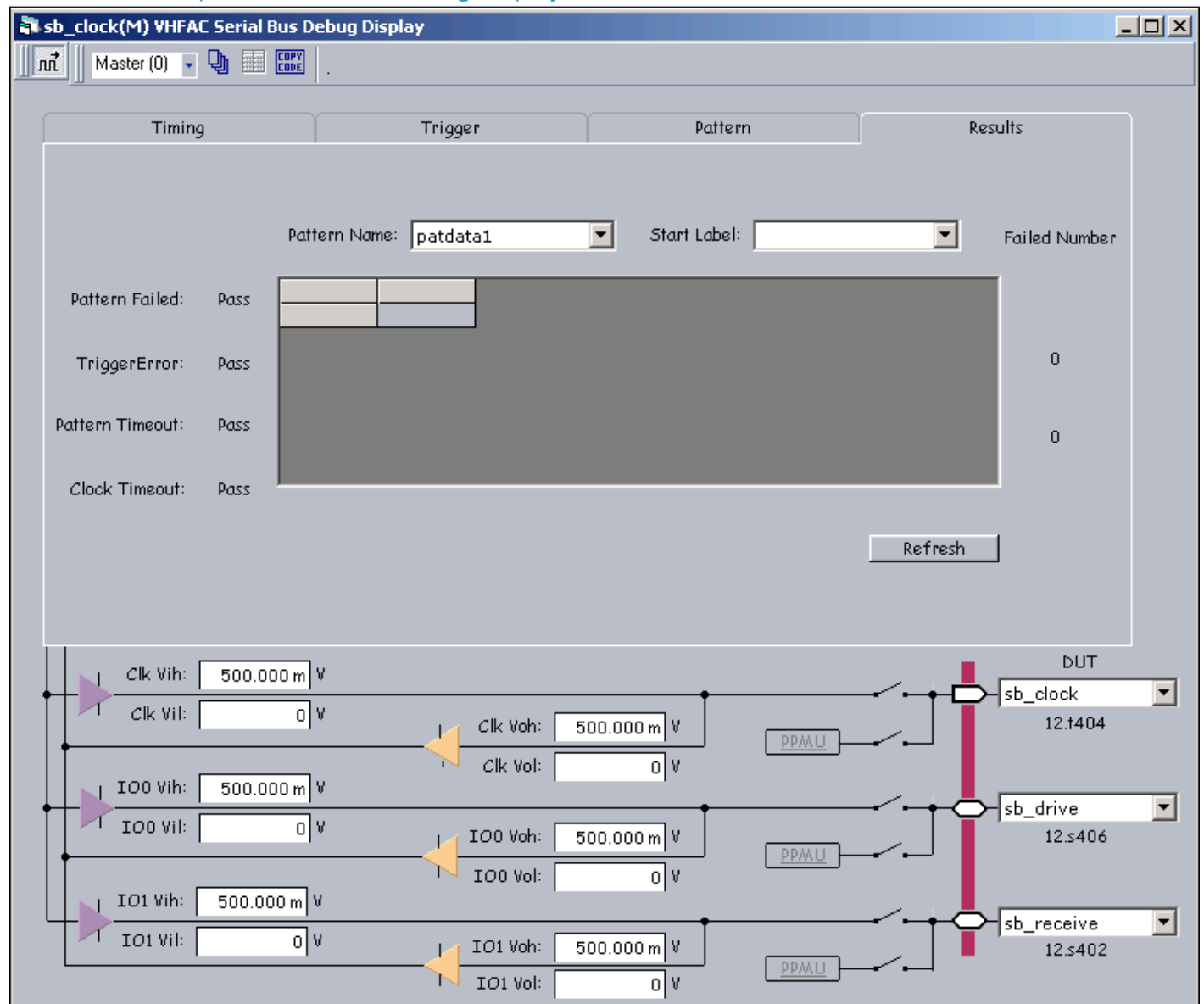
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VHFAC Serial Bus Debug Display : Debug Display Parameters and Buttons : VHFAC Serial Bus Results

VHFAC Serial Bus Results

The Results tab produces the following display:



The Serial Bus Schematic Results Tab

The controls of the Results tab are as follows:

VHFAC Serial Bus Schematic: Results Tab Controls

Parameters/ Buttons	Description
Pattern Name	This drop-down list displays the loaded pattern name. The result of the selected pattern can be displayed.
Start Label	This drop-down list displays only the start label which constitutes the selected pattern name. The start label for displaying a result is selected.
Failed Number	Displays the number of the failed pattern.
Pattern Failed	Displays the existence of pattern fail.
TriggerError	Displays the existence of trigger error.
Pattern Timeout	Displays the existence of pattern timeout.
Clock Timeout	Displays the existence of clock timeout.

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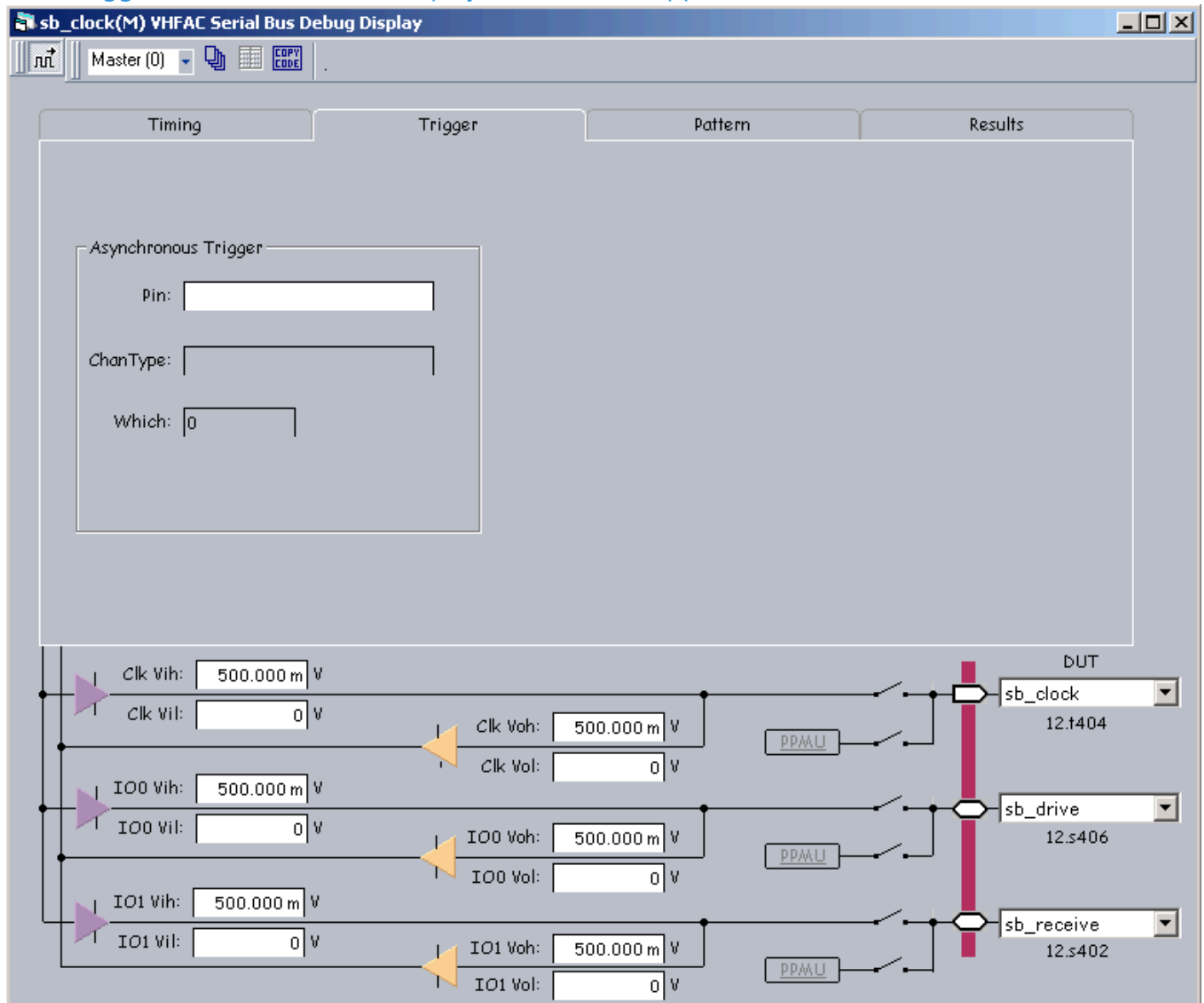
VhfacSerialBus_debug.4.7



VHFAC Serial Bus Debug Display : Debug Display Parameters and Buttons : VHFAC Serial Bus Trigger

VHFAC Serial Bus Trigger

If the Trigger tab is selected, the display shown below appears.



Serial Bus Debug Display Trigger Tab

The controls for the Trigger tab are:

VHFAC Serial Bus Schematic: Trigger Tab Controls

Parameters/ Buttons	Description
Pin	Specifies the pin whose signal is used to asynchronously trigger the instrument.
ChanType	This is the asynchronous trigger pin's channel type. The trigger channel type should match the asynchronous trigger pin channel type.
Which	A number identifying the trigger line used to asynchronously trigger the instrument.

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VhfacSerialBus_example.5.1

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VHFAC Serial Bus Examples

VHFAC Serial Bus Examples

This section provides full example code for the VHFAC Serial Bus instrument.
For more information, see:

- [VHFACSerialBus \(VBT Syntax Reference\)](#)
- [VHFAC Serial Bus Programming](#)

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VhfacSerialBus_example.5.2



[VHFAC Serial Bus Examples](#) : VBT Code Example

[VBT Code Example](#)

[Option Explicit](#)

Public Function SerialBusExample(ClkPin As String, DrivePin As String, ReceivePin As String)

Dim pldRetVal As New PinListData

' Set up Patterns (only on first run)

If TheExec.ExecutionCount < 1 Then

 ' Unload and remove all pattern data loaded in pattern memory, before

 ' defining new pattern data

 TheHdw.VHFACSerialBus.Pins(ClkPin).Patterns.Unload

 TheHdw.VHFACSerialBus.Pins(ClkPin).Patterns.Remove

 ' Define pattern data using ASCII format

 With TheHdw.VHFACSerialBus.Pins(ClkPin).Patterns("PatData1").Data

 .ClockLeadingEdge.Ascii = "0000 0000 0000 0000"

 .ClockTrailingEdge.Ascii = "1111 1111 1111 1111"

 .Drive0.Ascii = "0101 0101 1111 0000"

 .Receive1.Ascii = "LHLH LHLH HHHH LLLL"

 End With

 ' Load all defined patterns into the pattern memory.

 TheHdw.VHFACSerialBus.Pins(ClkPin).Patterns.Load

End If

```
' Connect the pins to the DUT
TheHdw.VHFACSerialBus.Pins(ClkPin).Connect
TheHdw.VHFACSerialBus.Pins(DrivePin).Connect
TheHdw.VHFACSerialBus.Pins(ReceivePin).Connect

' Set up the pin timing
' Clock pin
With TheHdw.VHFACSerialBus.Pins(ClkPin).Timing
    .AutoReprogram = True
    .OverrideRange.Autorange = True
    .Period = 0.000001          ' 1000ns
    .Clock.LeadingEdge = 0.0000001 ' 100ns
    .Clock.TrailingEdge = 0.0000006 ' 600ns
End With

' Drive pin
With TheHdw.VHFACSerialBus.Pins(DrivePin).Timing
    .DataDelay.Enable = True
    .DataDelay.Value = 0.0000002 ' 200ns
End With

' Receive pin
With TheHdw.VHFACSerialBus.Pins(ReceivePin).Timing
    .DataDelay.Enable = False
    .ReceiveStrobe = tIVHFACSerialBusReceiveStrobeTrailing
End With

' Pattern and Timing "time-outs"
With TheHdw.VHFACSerialBus.Pins(ClkPin).Timing
    .PatternTimeout.Enable = True
    .PatternTimeout.Value = 0.0005          ' 500us
    .Clock.TimeOut.Mode = tIVHFACSerialBusClockTimeoutModeLeading
    .Clock.TimeOut.Value = 0.000001        ' 1000ns
End With

' Set up pin levels
' Clock pin
```

```
With TheHdw.VHFACSerialBus.Pins(ClkPin).PinLevels
    .Vih.Value = 5#    ' 5.0V
    .Vil.Value = 0#    ' 0.0V
    .InitialDriveState = tlvHFACSerialBusInitialDriveStateOffLow
End With
```

```
' Drive pin
```

```
With TheHdw.VHFACSerialBus.Pins(DrivePin).PinLevels
    .Vih.Value = 5#    ' 5.0V
    .Vil.Value = 0#    ' 0.0V
    .InitialDriveState = tlvHFACSerialBusInitialDriveStateOffLow
End With
```

```
' Receive pin
```

```
With TheHdw.VHFACSerialBus.Pins(ReceivePin).PinLevels
    .Voh.Value = 2.7   ' 2.7V
    .Vol.Value = 2.2   ' 2.2V
    .InitialDriveState = tlvHFACSerialBusInitialDriveStateOffLow
End With
```

```
' Set up and start the pattern
```

```
With TheHdw.VHFACSerialBus.Pins(ClkPin).Patterns
    .SetStartPosition "PatData1"
    .StartSignal = tlvHFACSerialBusStartSignalByCPU
    .EnableStart
    .Start
End With
```

```
' Check the result of the pattern burst
```

```
' Confirm whether pattern fail occurred and save the value in PinListData.
```

```
pldRetVal = TheHdw.VHFACSerialBus.Pins(ClkPin).Results.PatternFailed
```

```
' Disconnect the pins from the DUT
```

```
TheHdw.VHFACSerialBus.Pins(ClkPin).Disconnect
TheHdw.VHFACSerialBus.Pins(DrivePin).Disconnect
TheHdw.VHFACSerialBus.Pins(ReceivePin).Disconnect
```

```
End Function
```

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VhfacSerialBus_prog.3.01

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VHFAC Serial Bus Programming

VHFAC Serial Bus Programming

This section describes programming of a VHFAC Serial Bus.

The following topics are available:

- VHFAC Serial Bus DataTool Programming
- Writing the VBT Code

For more information, see:

- VHFACSerialBus (VBT Syntax Reference)
- VHFAC Serial Bus Examples

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VHFAC Serial Bus Programming : VHFAC Serial Bus DataTool Programming

VHFAC Serial Bus DataTool Programming

This section describes how to program several DataTool worksheets to handle basic VHFAC Serial Bus parameters. It includes the following topics:

- Programming the Pin Map
- Programming the Channel Map

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VHFAC Serial Bus Programming : [VHFAC Serial Bus DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name.

An IG-XL pin map for the VHFAC Serial Bus instrument is shown below. Assign the VHFAC Serial Bus Clock channel and I/O channels the I/O pin type in the Pin Map sheet.

Pin Map

Group Name	Pin Name	Type	Comment
	sb_clock	I/O	VHFAC Serial Bus Clock
	sb_drive	I/O	VHFAC Serial Bus I/O 0
	sb_receive	I/O	VHFAC Serial Bus I/O 1

Pin Map Sheet

For more information, see [Pin Map Sheet](#) in the DataTool section.

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VhfacSerialBus_prog.3.04

VHFAC Serial Bus Programming : VHFAC Serial Bus DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. Assign the VHFAC Serial Bus Channels to VHFAC Serial Bus Clock, VHFAC Serial Bus I/O 0 or VHFAC Serial Bus I/O 1 channel types in a Channel Map sheet. The following examples show an UltraFLEX test system with two VHFAC Serial Bus instruments.

Channel Map

DIB ID:
View Mode:

Device Under Test		Tester Channel			
Pin Name	Package Pin	Type	Site 0	Site 1	Comment
sb_clock	1	VHFACSerialBusClock	22.sbclk	31.sbclk	VHFAC Serial Bus Clock
sb_drive	2	VHFACSerialBusIO0	22.sb0	31.sb0	VHFAC Serial Bus I/O 0
sb_receive	3	VHFACSerialBusIO1	22.sb1	31.sb1	VHFAC Serial Bus I/O 1

Channel Map Sheet, Two Instruments (Signal View)

This example shows two VHFAC Serial Bus instruments in Pogo view.

Channel Map

DIB ID:
View Mode:

Device Under Test		Tester Channel			
Pin Name	Package Pin	Type	Site 0	Site 1	Comment
sb_clock	1	VHFACSerialBusClock	22.d34	31.d34	VHFAC Serial Bus Clock
sb_drive	2	VHFACSerialBusIO0	22.c33	31.c33	VHFAC Serial Bus I/O 0
sb_receive	3	VHFACSerialBusIO1	22.d32	31.d32	VHFAC Serial Bus I/O 1

Channel Map Sheet, Two Instruments (Pogo View)

The following examples show one VHFAC Serial Bus instrument.

Channel Map					
DIB ID:		View Mode: Signal			
Device Under Test		Tester Channel			
Pin Name	Package Pin	Type	Site 0	Site 1	Comment
sb_clock	1	VHFACSerialBusClock	22.sbc1k	site0	VHFAC Serial Bus Clock
sb_drive	2	VHFACSerialBusIO0	22.sb0	site0	VHFAC Serial Bus I/O 0
sb_receive	3	VHFACSerialBusIO1	22.sb1	site0	VHFAC Serial Bus I/O 1

Channel Map Sheet, One Instrument (Signal View)

Channel Map					
DIB ID:		View Mode: Pogo			
Device Under Test		Tester Channel			
Pin Name	Package Pin	Type	Site 0	Site 1	Comment
sb_clock	1	VHFACSerialBusClock	22.d34	site0	VHFAC Serial Bus Clock
sb_drive	2	VHFACSerialBusIO0	22.c33	site0	VHFAC Serial Bus I/O 0
sb_receive	3	VHFACSerialBusIO1	22.d32	site0	VHFAC Serial Bus I/O 1

Channel Map Sheet, One Instrument (Pogo View)

For more information, see [Channel Map Sheet](#) in the DataTool section.

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VhfacSerialBus_prog.3.05

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[VHFAC Serial Bus Programming](#) : Writing the VBT Code

Writing the VBT Code

The following sections describe setting up VBT code for a typical test program:

- | | |
|---|---------------------------------------|
| • | Set Up Pattern Data |
| • | Connect DUT pins |
| • | Set Up Pin Timing |
| • | Set Up Pin Levels |
| • | Burst the Pattern |
| • | Check Pattern Results |

For more information on VHFAC Serial Bus VBT syntax, see [VHFACSerialBus](#) in the VBT Syntax Reference.

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VhfacSerialBus_prog.3.06

	
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VHFAC Serial Bus Programming : Writing the VBT Code : Set Up Pattern Data

Set Up Pattern Data

Pattern data for the VHFAC Serial bus is generated as ASCII strings of vector instructions for the clock and I/O pins of the instrument. Pattern related VBT syntax is described in [Pins.Patterns](#) in the VBT Syntax Reference.

This is an example of how to set up basic pattern data:

' Set up patterns (only on first run)

If TheExec.ExecutionCount < 1 Then

 ' Unload and remove all pattern data loaded in pattern memory,

 ' before defining new pattern data

 TheHdw.VHFACSerialBus.Pins("sb_clock").Patterns.Unload

 TheHdw.VHFACSerialBus.Pins("sb_clock").Patterns.Remove

 ' Define pattern data using ASCII format

 With TheHdw.VHFACSerialBus.Pins("sb_clock").Patterns("PatData1").Data

 .ClockLeadingEdge.Ascii = "0000 0000 0000 0000"

 .ClockTrailingEdge.Ascii = "1111 1111 1111 1111"

 .Drive0.Ascii = "0101 0101 1111 0000"

 .Receive1.Ascii = "LHLH LHLH HHHH LLLL"

 End With

 ' Load all defined patterns into the pattern memory.

 TheHdw.VHFACSerialBus.Pins("sb_clock").Patterns.Load

End If

If the programmed pattern data is to remain static over multiple runs of the program, it can be created only once during the first run of the program for a more efficient program. This is done in the above code example by placing the pattern setup code inside the `If TheExec.ExecutionCount < 1 Then ... End If` syntax.

In addition to creating simple pattern data, as shown above, you can also define [Pattern Groups](#) and [Pattern Sets](#). VBT syntax for [Pattern Groups](#) and [Pattern Sets](#) is described in [Pins.Patterns.Item.PatternGroups](#) and [Pins.Patterns.Item.PatternSets](#) in the VBT Syntax Reference.

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VhfacSerialBus_prog.3.07

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VHFAC Serial Bus Programming : Writing the VBT Code : Connect DUT pins

Connect DUT pins

Connect and disconnect pins to and from a DUT using the following example VBT syntax:

' Connect the pin to the DUT

TheHdw.VHFACSerialBus.Pins("sb_clock").Connect

' Disconnect the pin from the DUT

TheHdw.VHFACSerialBus.Pins("sb_clock").Disconnect

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VHFAC Serial Bus Programming : Writing the VBT Code : Set Up Pin Timing

Set Up Pin Timing

You can program the desired timing for the clock and I/O pins of the instrument. Pin timing related VBT syntax is described in [Pins.Timing](#) in the VBT Syntax Reference. In addition, pin and pattern time-out related VBT syntax is described [TimeOut](#).

This is a basic example of programming pin timing and pin/pattern timeout:

' Set up the pin timing

' Clock pin

With TheHdw.VHFACSerialBus.Pins("sb_clock").Timing

.AutoReprogram = True

.OverrideRange.Autorange = True

.Period = 0.000001 ' 1000ns

.Clock.LeadingEdge = 0.0000001 ' 100ns

.Clock.TrailingEdge = 0.0000006 ' 600ns

End With

' Drive pin

With TheHdw.VHFACSerialBus.Pins("sb_drive").Timing

.DataDelay.Enable = True

.DataDelay.Value = 0.0000002 ' 200ns

End With

' Receive pin

With TheHdw.VHFACSerialBus.Pins("sb_receive").Timing

.DataDelay.Enable = False

.ReceiveStrobe = tIVHFACSerialBusReceiveStrobeTrailing

End With

' Pattern and Timing "time-outs"

With TheHdw.VHFACSerialBus.Pins("sb_clock").Timing

```
.PatternTimeout.Enable = True
.PatternTimeout.Value = 0.0005    ' 500us
.Clock.TimeOut.Mode = tlVHFACSerialBusClockTimeoutModeLeading
.Clock.TimeOut.Value = 0.000001    ' 1000ns
```

End With

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VhfacSerialBus_prog.3.09

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VHFAC Serial Bus Programming : Writing the VBT Code : Set Up Pin Levels

Set Up Pin Levels

You can program the desired levels for the clock and I/O pins of the instrument. Pin levels related VBT syntax is described in [Pins.PinLevels](#) in the VBT Syntax Reference. This is a basic example of programming pin levels:

' Set up pin levels

' Clock pin

With TheHdw.VHFACSerialBus.Pins("sb_clock").PinLevels

.Vih.Value = 5# ' 5.0V

.Vil.Value = 0# ' 0.0V

.InitialDriveState = tIVHFACSerialBusInitialDriveStateOffLow

End With

' Drive pin

With TheHdw.VHFACSerialBus.Pins("sb_drive").PinLevels

.Vih.Value = 5# ' 5.0V

.Vil.Value = 0# ' 0.0V

.InitialDriveState = tIVHFACSerialBusInitialDriveStateOffLow

End With

' Receive pin

With TheHdw.VHFACSerialBus.Pins("sb_receive").PinLevels

.Voh.Value = 2.7 ' 2.7V

.Vol.Value = 2.2 ' 2.2V

.InitialDriveState = tIVHFACSerialBusInitialDriveStateOffLow

End With

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VHFAC Serial Bus Programming : Writing the VBT Code : Burst the Pattern

Burst the Pattern

You are now ready to set up the instrument to burst a specific pattern. Pattern burst setup related VBT syntax is described in Pins.Patterns in the VBT Syntax Reference. This is a basic example of setting up and bursting a pattern:

' Set up and burst the pattern

With TheHdw.VHFACSerialBus.Pins("sb_clock").Patterns

```
.SetStartPosition "PatData1"  
.StartSignal = tIVHFACSerialBusStartSignalByCPU  
.EnableStart  
.Start
```

End With

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VHFAC Serial Bus Programming : Writing the VBT Code : Check Pattern Results

Check Pattern Results

Finally, you are ready to write code to check the results of the pattern burst. Pattern result related VBT syntax is described in Pins.Results in the VBT Syntax Reference. This is a basic example of checking whether the last run pattern passed all vectors or failed any number of vectors. The PinListData has a True for each site where a pattern failed:

```
Dim pldRetVal As New PinListData
    ' Check the result of the pattern burst
    ' Confirm whether pattern fail occurred and save the value in PinListData.
    pldRetVal = TheHdw.VHFACSerialBus.Pins(ClkPin).Results.PatternFailed
```

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VhfacSerialBus_theory

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VHFAC Serial Bus Theory of Operation

VHFAC Serial Bus Theory of Operation

The VHFAC Serial Bus instrument has three digital channels, one external trigger line, and one condition line.

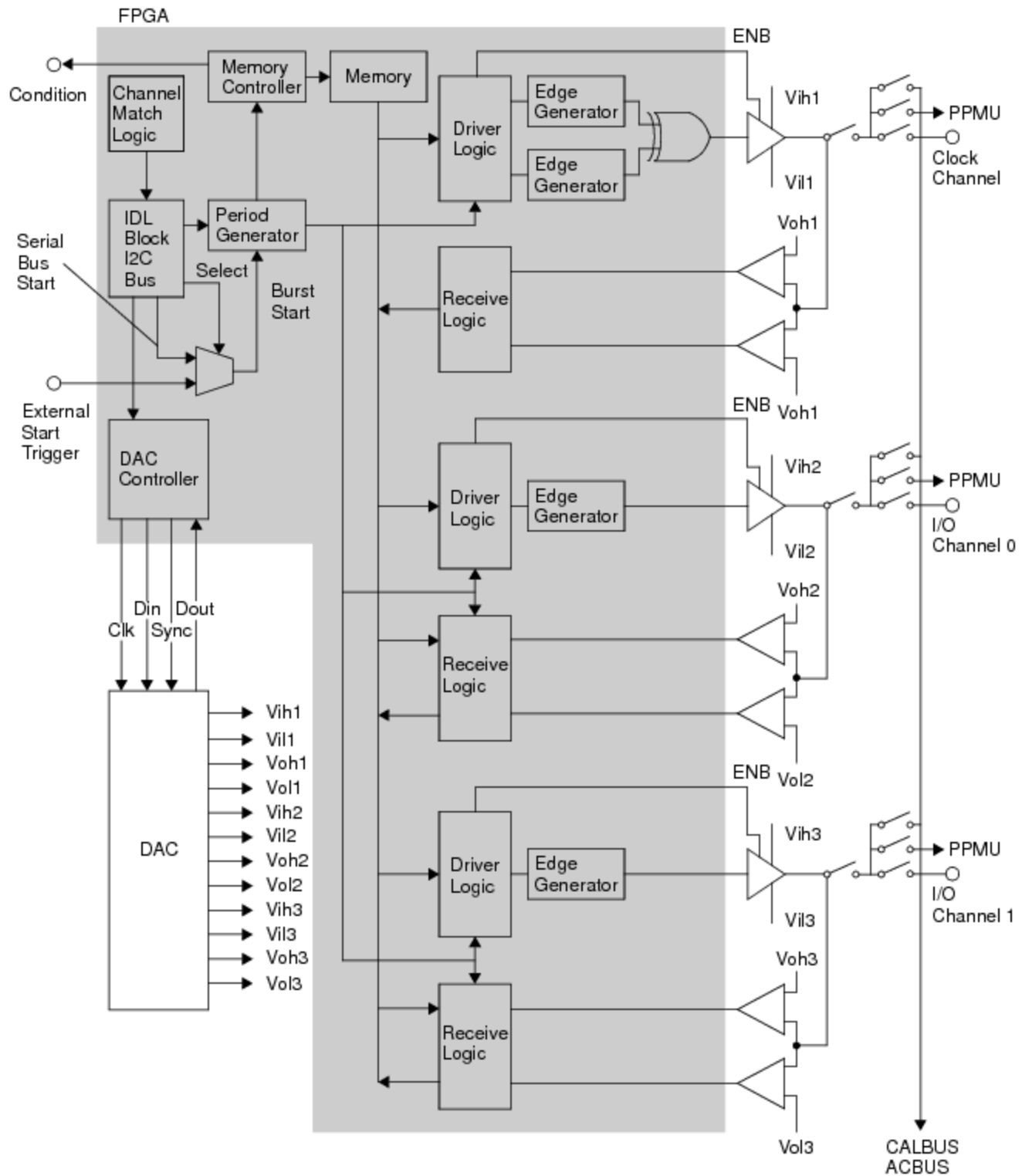
The three digital channels consist of one clock channel and two I/O channels. Each channel has tristate drivers, dual level receive comparators, and 64K of pattern RAM. The I/O lines are used to write data into and read data out of the DUT.

The start signal for starting patterns can originate from:

- | | |
|---|---|
| • | Test program (CPU start) |
| • | A trigger source connected to a external trigger line |

The trigger line is used when starting a pattern from a trigger source like VHFAC Source, Capture, or TMU.

The condition lines are used to indicate that the VHFAC Serial Bus has halted at the end of a pattern, pattern group, or pattern set. This can be used to synchronize to other signal sources. For information on VHFAC Serial Bus instrument performance specifications and standards, see VHFAC Specifications.



VHFAC Serial Bus Instrument Hardware Block Diagram

